

Muharram Syah

Jakarta, Indonesia • +6281361984557 • muharramsyah19@gmail.com • in/muharram-syah

PROFESSIONAL SUMMARY

Machine Learning Engineer expert in **Computer Vision or Vision Language Model (VLM)**, NLP, and Generative AI. Specialized in deploying and maintaining large-scale models using vLLM for optimized throughput and latency. Proven expertise in fine-tuning, quantization, and building **MLOps** workflows on AWS/GCP to bridge the gap between AI research and production-scale applications.

SKILLS AND TECHNOLOGIES

- Programming Languages: Python, SQL, JavaScript, TypeScript
- ML Framework: PyTorch, TensorFlow, Hugging Face, YOLO, MaskRCNN, NVIDIA DeepStream, TensorRT, Langchain, vLLM
- Specialization: Computer Vision, LLMs, Generative AI, Model Optimization (SFT, pruning, distillation, quantization), Prompt Engineering, LLM Distillation, AI Multi-Agent, Retrieval-Augmented Generation
- Cloud / MLOps: AWS SageMaker, AWS Lambda, Azure ML, Docker, Kubernetes, CI/CD, RESTful APIs, AWS ECR, Dify, N8N Automation , Claude Code SDK, Unsloth (LLM Distillation Framework)

PROFESSIONAL EXPERIENCE

AppFuxion Consulting Indonesia – Senior Machine Learning Engineer Sep 2025 – Present, Jakarta, Indonesia

- Developed a high-performance Document Extraction engine powered by Vision Language Models (VLM) to transform unstructured visual data into a structured Knowledge Base for the AGIS Platform. Engineered specialized pipelines for parsing complex layouts, tables, and charts, significantly enhancing the platform's multi-modal data ingestion capabilities.
- Streamlined production deployment of multi-modal models using vLLM, achieving significant improvements in token throughput and latency reduction. Managed server-side orchestration to ensure seamless model availability and scalability under high-concurrency production workloads.
- Spearheaded the development of a Document Intelligence Processing tool during the POC phase, digitizing legacy paper-based workflows into structured, machine-readable data. Architected the integration of OCR and VLM-based extraction to handle complex document schemas with high precision.

Petronas Digital Sdn Bhd – Machine Learning Engineer Aug 2024 – Jan 2025, Kuala Lumpur, Malaysia

- Spearheaded the optimization of AWS SageMaker systems for real-time inference, achieving an 81.6% reduction in operational and cloud costs through innovative strategies.
- Led the deployment of AI models on AWS cloud infrastructure, ensuring seamless integration and scalability.
- Engineer advanced image analytics modules with YOLOv8, automating composite repair assessments and enhancing inspection accuracy.

Esri Indonesia – Engineering Lead Feb 2023 - May 2024, Jakarta, Indonesia

- Directed a team of 6 engineers, resolving technical challenges and improving team efficiency.
- Architected and implemented microservices architectures integrated with secure and high-performance internal API gateways.
- Mentored junior data scientists and engineers, fostering a culture of continuous learning and innovation.

Esri Indonesia – Data Scientist May 2020 – Feb 2023, Jakarta, Indonesia

- Built a real-time traffic anomaly detection system using YOLOv4 and NVIDIA DeepStream, enabling faster incident response and improved public safety outcomes.
- Developed an NLP pipeline for disaster monitoring from X.com (forming Twitter), providing insurers with faster risk assessment and reducing claim processing time through dashboard monitoring.
- Automated palm fruit grading with MaskRCNN and TensorFlow Lite, improving harvest efficiency, reducing waste, and ensuring higher-quality palm oil production.
- Designed a wellbore failure detection model using Computer Vision on rock cuttings, enabling early drilling risk identification and reducing costly broken equipment in oil & gas operations.

Telkom Indonesia – AI Scientist (Intern) March 2018 – March 2019, Jakarta, Indonesia

- Deployed object detection with YOLO across 16 CCTV systems in Rest Area Management to improve traffic flow on Highways in Indonesia.
- Built a nationwide real-time fuel operations dashboard, providing instant visibility across gas stations and enabling faster decision-making and operational efficiency.
- Integrated smart office absence tracking with CCTV surveillance streamlines attendance monitoring, reducing manual errors and enhancing workplace security.

PROJECTS

- Advanced Generative Intelligent System (AGIS) 2025** - Developed and deployed AGIS, an enterprise-grade AI ecosystem designed to capture and synthesize organizational intelligence. Leveraged Multi-Agent orchestration to automate evolving business workflows, significantly reducing manual data processing and enhancing operational agility.
- Call Center Intelligence (2025)** - Designed and implemented an AI-powered customer service analytics system with speech-to-text diarization, sentiment detection, enabling faster issue resolution, improved customer satisfaction, and data-driven agent coaching.
- SQL Intelligence (2025)** - Developed an AI-powered SQL Intelligence system using LLMs and AI Agents (Claude Code) to automatically convert DDL, and Data Quality pipelines, accelerating database migration and reducing manual effort.
- LLM Distillation (2025)** - Fine-tuned Qwen-14B with DeepSeek R1 to create a lightweight, cost-efficient recommendation system, reducing inference costs on AWS Bedrock or SageMaker while maintaining high accuracy.
- CORIMS (2024)** – Engineered an AI-driven predictive maintenance platform using YOLOv8, enabling operators to automatically detect composite repair issues and streamline inspection workflows on AWS.
- Traffic Anomaly Detection (2022)** – Deployed a real-time traffic incident risk prediction system with YOLOv4 and NVIDIA DeepStream SDK, improving public safety response and reducing incident handling time.
- Forest Fire Detection (2021)** – Built an Early warning system with Sentinel Hub and PyTorch, enabling authorities to identify wildfire risks in real time and take preventive action.

EDUCATION

Bachelor of Technology in Computer Science – Binus University (2019)

Thesis Statement: **Vehicle Counting and Classification using YOLO**

PROFESSIONAL CERTIFICATE

- PCAP – Certified Associate in Python Programming (OpenEDG Python Institute)
- ArcGIS Developer Foundation 2201 (Esri)
- TensorFlow Developer Certificate (TensorFlow Developer Program)
- Microsoft Azure: AI Fundamental Certificate (Microsoft)
- Microsoft Azure: AI Engineer Associate (Microsoft)

ADDITIONAL HIGHLIGHT

- Strong experience with AI model integration in real-world platforms, including enterprise and industry AI applications.
- Continuous learner, actively following the latest AI/ML advancements and implementing state-of-the-art AI techniques